

DRIVER DROWSINESS DETECTION SYSTEM USING RASHBERRY PI

Team members

M.Lavanya

T.Anujasri

S.Dharshini

S.Preethika

Problem statement

Road accidents caused by driver drowsiness are a major safety concern worldwide. Drivers often become tired or sleepy during long-distance or night-time driving, which reduces their attention, slows their reaction time, and increases the risk of accidents. Most vehicles do not have an affordable system to detect driver drowsiness and provide timely alerts. Existing solutions are often expensive, require specialized hardware, or are difficult to implement in low-cost vehicles. Therefore, there is a need for a cost-effective, real-time driver drowsiness detection system that can continuously monitor the driver's eye movements and alert the driver immediately when signs of drowsiness are detected. This project aims to develop such a system using a Raspberry Pi, camera, and image processing techniques.

Abstract

Driver drowsiness is a critical factor contributing to road accidents, posing a significant threat to road safety worldwide. To mitigate this issue, a Driver Drowsiness Driver Detection System (DDDS) using Raspberry Pi has been developed to provide a cost effective, efficient, and real-time solution. This system employs various sensor technologies, such as a camera for facial recognition, infrared sensors for eye-tracking, and a heart rate sensor to monitor the driver's physiological state.

By utilizing machine learning algorithms, the system can accurately detect signs of drowsiness, such as blinking rate, yawning frequency, and head movement, thus alerting the driver to take necessary action before fatigue-related accidents occur. The proposed system is based on a Raspberry Pi, chosen for its affordability, compactness, and ease of integration with peripheral sensors.

The camera captures the driver's facial features, which are then analysed using computer vision techniques to identify eye movements and facial expressions indicative of drowsiness. The heart rate sensor further supplements this data by monitoring physiological signs of fatigue. When drowsiness is detected, the system triggers an alarm, such as a sound or vibration, to warn the driver and ensure immediate attention.

The paper explores the design and implementation of the system, detailing the integration of hardware components and the development of the software architecture. Various algorithms, including Haar cascades and deep learning-based models, are employed for real-time drowsiness detection.

Furthermore, the system's performance is evaluated based on multiple test scenarios, focusing on detection accuracy, response time, and reliability under different lighting and environmental conditions. In conclusion, the proposed Driver Drowsiness Detection System offers a promising approach to enhancing road safety by leveraging the capabilities of Raspberry Pi and modern sensor technologies.

The system is scalable, adaptable to various vehicles, and provides a low-cost alternative to existing commercial solutions, making it an ideal candidate for widespread implementation in both private and commercial transportation. **Keywords:** Face Detection, Python, Open CV, Raspberry Iddlib

Objectives

The main objective of this project is to design and develop a Driver Drowsiness Detection System using Raspberry Pi that can monitor the driver's alertness in real time and help prevent road accidents caused by fatigue.

The specific objectives of this project are:

1. To develop a low-cost and efficient driver drowsiness detection system using Raspberry Pi.
2. To continuously monitor the driver's facial features, especially eye movements, using a camera.
3. To detect signs of drowsiness such as prolonged eye closure, yawning, and head movement using computer vision techniques.
4. To provide an immediate audio alert whenever driver fatigue is detected, helping the driver regain attention.
5. To improve road safety by reducing accidents caused by driver drowsiness.
6. To create a portable, reliable, and user-friendly system that can be used in both personal and commercial vehicles.
7. To provide a foundation for future enhancements such as Artificial Intelligence (AI), Internet of Things (IoT), GPS tracking, and cloud-based monitoring.
- To Design and Development of the system to detect drowsiness of the driver

- To alert drivers while driving the vehicle if they are feeling sleepy

This research aims at developing an IoT surveillance system for vehicular drivers using a smartphone to detect and alert the driver when he or she exhibits drowsy behaviour.

We reviewed other works related to detecting drowsiness in drivers, the symptoms of drowsiness in drivers, and determined the technical parameters and algorithms that are needed to process signals of the state of drowsiness.

Scope of the project

Moving forward, there are some things we are able to do to further improve our results and fine tune the models. First, we want to include distance between the facial landmarks to account for any movement by the topic within the video. Second, we would like to update parameters with our more complex models so as to attain better results. Third, we'd prefer to collect our own training data from a bigger sample of participants while including new distinct signals of drowsiness.

To enhance system reliability, future improvements can focus on:

- Infrared (IR) camera for better performance in low light conditions.
- Deep learning based facial tracking to improve accuracy under varying head positions.
- IoT integration for real time alerts to emergency contacts.
- Multi sensor fusion, combining heart rate and eye movement data for more precise detection.
- Adding automatic object detection.
- Controlling the acceleration of the vehicle.
- Alerting surrounding vehicles about drivers drowsiness.
- Enhancing security using smart sensors.
- Integrating GPS with the device, which may notify nearby hospitals and police stations with an accident occurs.

Expected outcomes

The research suggests driver behaviour-based drowsiness and intrusion detection. The system's job is to locate landmarks from images while the driver is driving and send these to a trained model to identify the drive state. By reading the collection of data showing signs of drowsiness, the driver will be warned and instructed to halt the vehicle to prevent driving mishaps.

Advanced driver-assistance systems, such as drowsiness detection of drowsy driving and mobile apps, will integrate it. However, even if we look bright, there is significant room for improvement. The driving portion of the research will focus on determining the driver's distraction and drowsiness. The created driver drowsiness system is able to quickly identify drowsy driving behaviours.

The drowsiness detection system, which was created based on the driver's eye closure, is able to distinguish between regular eye blinking and drowsiness and can identify drowsiness while driving. The suggested technique can stop accidents brought on by drivers who are drowsy. Several image processing methods are used to gather information about the position of the head and eyes.

The monitoring system can determine if the eyes are open or closed. When the eyes have been closed for too long, a warning signal is issued. Processing judges the driver's alertness level on the basis of continuous eye closures.

Literature survey

Based on the literature survey Based on the literature survey in your screenshots, here is a proper table (box format) that you can directly insert into your project report.

AUTHOR	CONTRIBUTION	LIMITATION
Anil kumar biswal et al.	Developed a raspberry pi 3 model b and pi camera based driver drowsiness detection system using facial landmark detection.The system provides voice and e-mail alerts.	Depends on internet connectivity for e-mail alerts and has limited real-time emergency response.
Jabina m.et al	Developed a driver safety system using an alcohol sensor,GSM module, and automatic vehicle control to prevent accidents.	Focuses mainly on alcohol detection rather than driver drowsiness.
R.Rajashekhar Reddy&Dr.Padmaja Pulicherla	Used facial features such as eye aspect, mouth opening, and nose reference to detect	Performance is affected by poor lighting and face occlusion.

	drowsiness and warn the driver.	
Badiuzzaman Bin Baharu	Proposed a PERCLOS-based eye closure monitoring system for detecting driver fatigue and generating alerts.	Detects only eye closure and is sensitive to illumination changes.
Pratyush Agarwal&Rizal Sharma	Used pupil center detection and iris localization to identify drowsiness and trigger an alarm.	Requires high computational power and accurate eye localization.
Tian et al.	Developed an eye-state detection system using the YCbCr color model and facial feature extraction.	Accuracy decreases with head movement and varying lighting conditions.
Chellappa et al.	Combined camera, alcohol sensor, and heart-rate sensor with machine learning to monitor driver condition in real time.	Increases system complexity and hardware cost due to multiple sensors.
Patel et al.	Proposed an IoT-based driver drowsiness and alcohol detection system with vehicle tracking and emergency notification.	Depends on iot/GSM connectivity and requires internet access.
Uma et al.	Applied machine learning and IoT to detect drowsiness using eye analysis and MQ6 alcohol sensing.	Requires continuous sensor monitoring and additional hardware.
Kiran et al.	Integrated Raspberry Pi, Pi Camera, alcohol sensor, GPS, seat belt sensor, and buzzer for driver safety.	Higher implementation cost and increased hardware complexity.
Pooja D.C. et al.	Implemented a Raspberry Pi and OpenCV-based real-time drowsiness detection system using Haar Cascade face and eye	Haar Cascade accuracy decreases under poor lighting and different face orientations.

	detection.	
Proposed System	Uses Raspberry Pi, Pi Camera, OpenCV, Dlib facial landmarks, Eye Aspect Ratio(EAR), Mouth Aspect Ratio(MAR). and buzzer alarm for real-time driver drowsiness detection.	Requires proper camera positioning and adequate lighting for best performance.

Existing solution analysis

Driver drowsiness detection has become an important area of research due to the increasing number of road accidents caused by fatigue. Existing systems use different techniques such as camera-based face detection, eye-blink monitoring, infrared sensors, wearable devices, and vehicle behavior analysis to identify signs of driver drowsiness. Many modern systems apply computer vision and machine learning algorithms to monitor the driver's facial expressions, eye movements, blinking patterns, yawning, and head position in real time. When drowsiness is detected, the system generates an alert to warn the driver.

Despite their effectiveness, existing solutions have several drawbacks. Most commercial driver monitoring systems are costly and are available only in premium vehicles. Wearable sensor-based systems may cause discomfort to drivers during long-distance travel. Vehicle-based monitoring methods, such as steering wheel movement and lane departure detection, may identify fatigue only after the driver's performance has already deteriorated. In addition, some camera-based systems show reduced accuracy in poor lighting conditions, when the driver wears spectacles or sunglasses, or when facial features are partially blocked.

To overcome these limitations, the proposed Driver Drowsiness Detection System uses a Raspberry Pi with a camera, OpenCV, Dlib, and Python to monitor the driver's eyes and facial features continuously. The system detects signs of drowsiness such as prolonged eye closure, frequent yawning, and head nodding. Once fatigue is detected, an immediate alarm is activated to alert the driver. The proposed solution is low-cost, portable, easy to install, and suitable for both private and commercial vehicles, making it an effective and affordable alternative to existing systems.

Existing driver drowsiness detection systems use a variety of technologies to monitor the driver's alertness and reduce fatigue-related road accidents. Most systems rely on camera-based facial recognition, infrared eye-tracking sensors, heart rate monitoring, and computer vision techniques to detect signs of drowsiness such as eye closure, blinking rate, yawning, and head movement. Some advanced systems also employ machine learning and deep learning algorithms, including Haar Cascade classifiers and facial landmark detection using Dlib, to improve detection accuracy. When drowsiness is identified, these systems generate audio or vibration alerts to warn the driver and help prevent potential accidents.

Although these existing solutions provide good detection accuracy, they have several limitations. Many commercial systems are expensive and available only in premium vehicles, making them unsuitable for widespread adoption. Wearable sensor-based methods may be uncomfortable for drivers, while some camera-based systems are affected by poor lighting conditions, spectacles, or changes in head position. To address these issues, the proposed Driver Drowsiness Detection System uses Raspberry Pi as a compact and low-cost processing platform integrated with a camera, OpenCV, Python, and Dlib. The system performs real-time monitoring of the driver's facial features and provides immediate alerts when drowsiness is detected, offering an affordable, reliable, and practical solution for both personal and commercial vehicles.

Gap analysis

Existing driver drowsiness detection systems provide effective monitoring using technologies such as computer vision, infrared sensors, wearable devices, and machine learning algorithms. However, many of these systems are expensive, require specialized hardware, or are available only in premium vehicles. Some wearable solutions are inconvenient for drivers, while vehicle behavior-based methods detect fatigue only after driving performance has been affected. In addition, the accuracy of certain camera-based systems decreases under poor lighting conditions, when drivers wear spectacles or sunglasses, or due to variations in facial orientation.

The proposed Driver Drowsiness Detection System addresses these gaps by using Raspberry Pi as a low-cost, compact, and energy-efficient platform integrated with a camera, OpenCV, Python, and Dlib. The system performs continuous real-time monitoring of eye movements, facial landmarks, yawning, and head position to detect drowsiness at an early stage. When fatigue is identified, an immediate alarm is generated to alert the driver. This approach provides an affordable, portable, and reliable solution that can be easily deployed in both personal and commercial vehicles while maintaining good detection accuracy.

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